

FinTrust Digital Bank — Week 3 Data Analytics

Advanced Business Intelligence & Dashboard Development

Week 3 builds directly on Week 2 through deeper channel/type/value analysis, outlier concentration, amount bands, hourly analysis, relationship-integrity controls, finding validation and management recommendations.

Improvement Areas

G1 — Customer-level transaction analysis is unreliable Evidence: 32 of 33 transaction Customer_IDs are unmatched. Improvement: Add relationship-integrity validation and prevent unsupported customer-level conclusions.

G2 — Week 2 dashboard was mainly descriptive Evidence: Week 2 established KPI/channel/type visuals but limited deeper validation. Improvement: Add value-share, amount-band, outlier-concentration and hourly analyses plus validation panels.

G3 — Some requested dimensions are unavailable Evidence: Transaction_Status and Risk_Review_Flag are absent from the transaction export. Improvement: Make data availability explicit and exclude unsupported KPIs from calculated results.

Validated Findings

V1. Mobile App is most-used and highest-value

Evidence: 14/33 transactions; ■493,341.73; 51.4% of sample value

Validation: Validated

Meaning: Both volume and value rank Mobile App first.

Action: Monitor mobile-channel experience; validate against broader data before operational decisions.

V2. Everyday is largest visible customer segment

Evidence: 17/33 customers (51.5%)

Validation: Validated

Meaning: Customer segment counts confirm Everyday is largest.

Action: Explore segment strategy only after reliable transaction-to-customer mapping.

V3. Transaction values are right-skewed

Evidence: Mean ■29,066.07 vs median ■6,702.14

Validation: Validated

Meaning: Five IQR outliers represent 71.4% of sample value.

Action: Use median/percentiles with mean and validate high-value records.

V4. Cash Withdrawal contributes the most value

Evidence: ■395,660.05; 41.2% of sample value

Validation: Validated

Meaning: Cash Withdrawal ranks first by total value.

Action: Review withdrawal patterns when complete status/risk fields are available.

V5. Customer/transaction relationship is unreliable

Evidence: 1/33 matched; 32/33 unmatched

Validation: Validated

Meaning: Customer-level attribution is not reliable.

Action: Validate authoritative source-key mapping before customer-level behavioural analysis.

Management Recommendations

R1. Mobile App leads volume and value

Evidence: 14 transactions; ■493,341.73

Implication: Digital channel is central in the sample

Action: Prioritize mobile-channel monitoring and experience review; validate on a larger period.

R2. Large transactions dominate value

Evidence: 5 outliers = 71.4% of value

Implication: Averages can be distorted by a small number of large records

Action: Report median/value concentration and validate high-value records.

R3. Cash Withdrawal is the largest value-driving type

Evidence: ■395,660.05 / 41.2%

Implication: Withdrawal activity materially influences sample value

Action: Review withdrawal patterns and channel distribution when complete fields are available.

R4. Customer-level behaviour cannot be attributed reliably

Evidence: 32/33 transaction IDs unmatched

Implication: Segment-to-transaction analysis could mislead

Action: Resolve source-key mapping before customer-level KPIs.

R5. Time coverage is narrow

Evidence: All transactions occur 2026-01-01 00:00–05:45

Implication: Hourly patterns are sample-specific

Action: Request broader date/time coverage before operational trend decisions.

Testing & Refinement

Week 2 outputs were independently recalculated in SQL and Python. Deeper analysis showed that value concentration and relationship integrity needed stronger visibility. Week 3 added value-share, amount-band, outlier, cumulative-value and hourly outputs plus a data-quality panel. Re-testing reconciled core totals and supported the five major findings, while qualifying hourly interpretation because of the narrow sample window.

Limitations

Transaction_Status and Risk_Review_Flag are absent. No unsupported values are fabricated. 32/33

Customer_IDs are unmatched, so customer-level transaction segmentation is withheld. The transaction sample covers only 2026-01-01 00:00–05:45.

Week 4 Readiness

Finalise the native Power BI Desktop report, test KPIs and visuals, refine presentation, document limitations and prepare the final demonstration/submission.